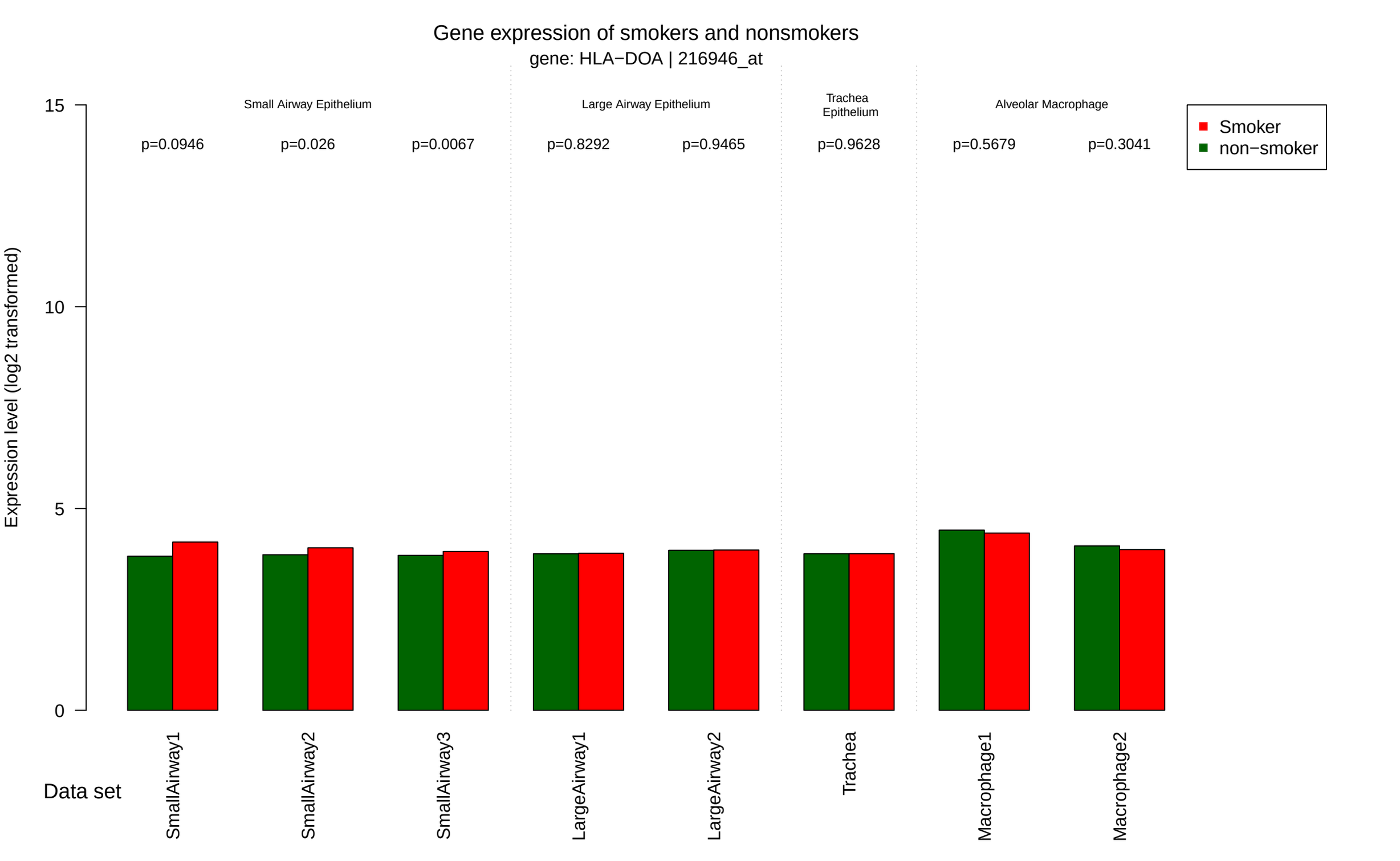
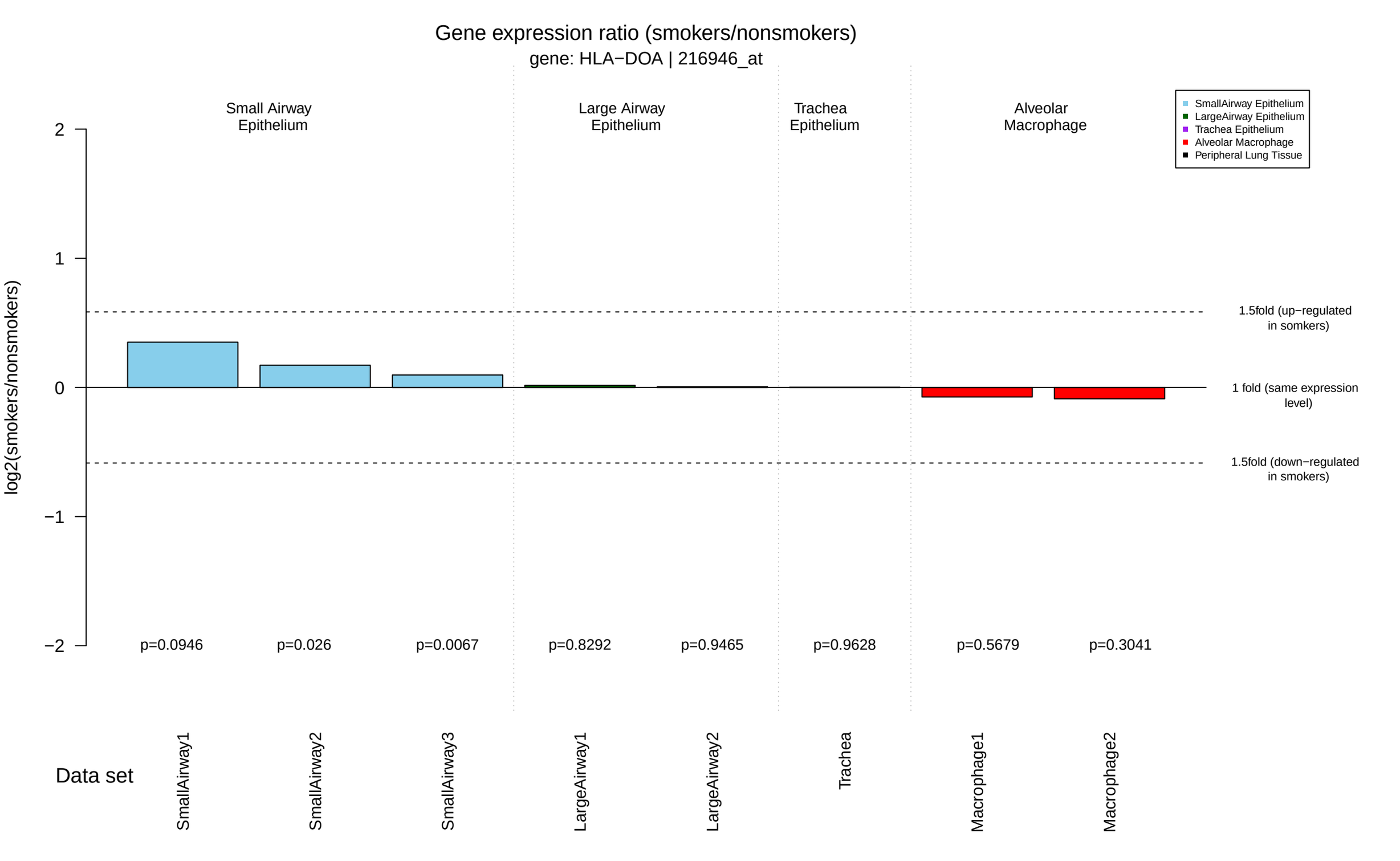
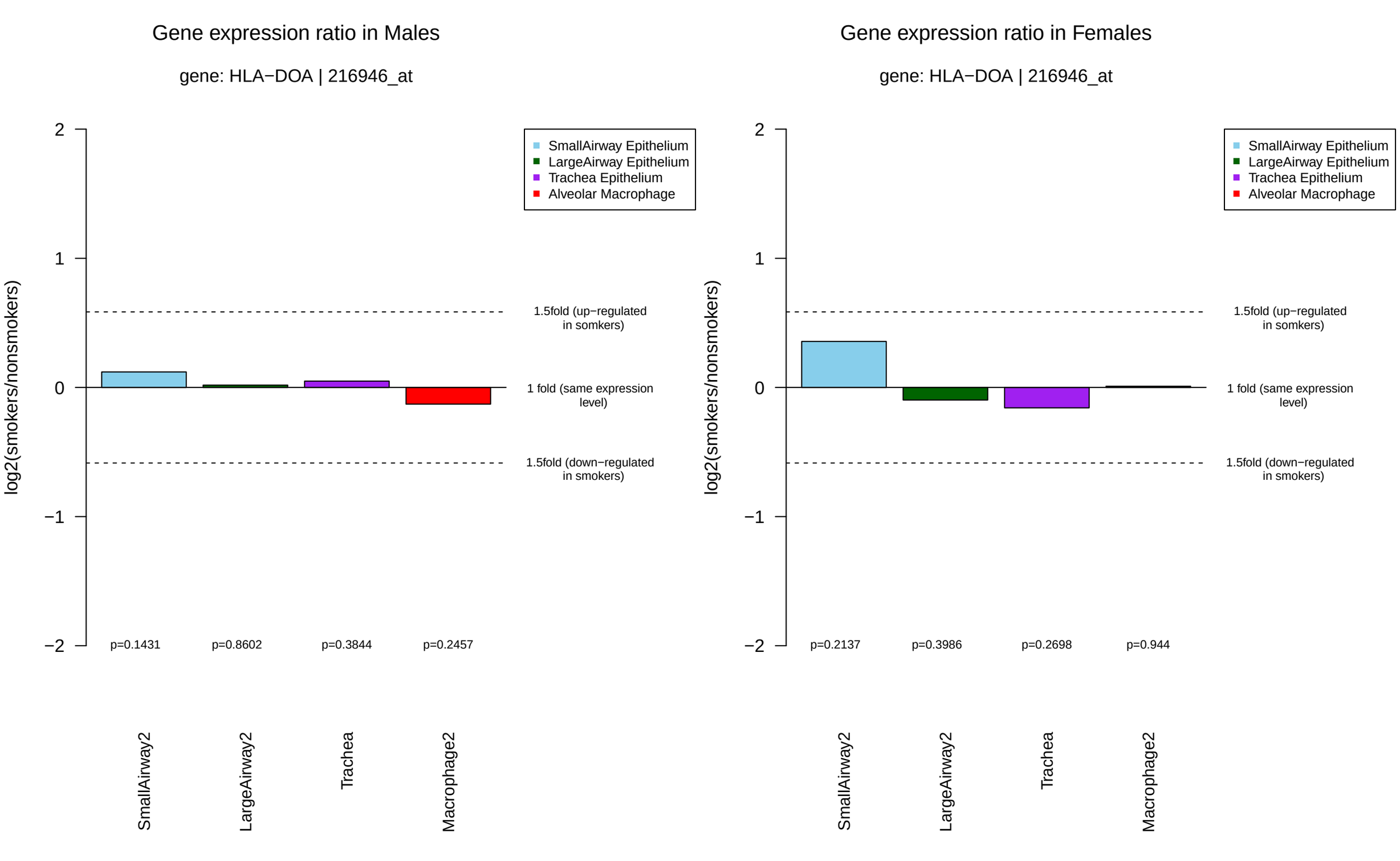
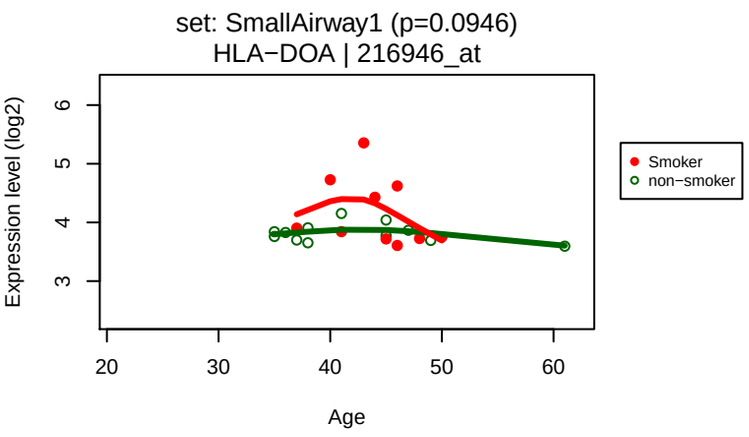
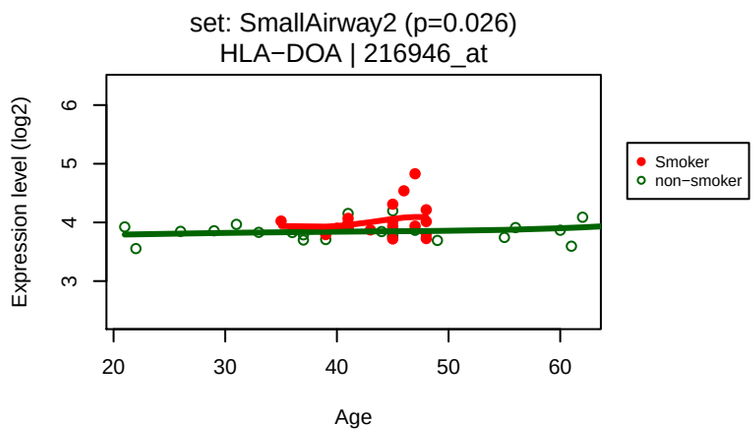
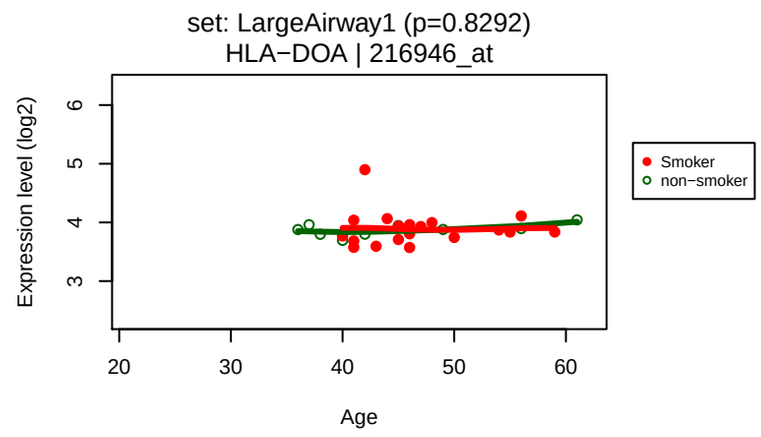

Gene expression of smokers and nonsmokers
gene: HLA−DOA | 216946_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0946
p=0.026
p=0.0067
p=0.8292
p=0.9465
p=0.9628
p=0.5679
p=0.3041
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

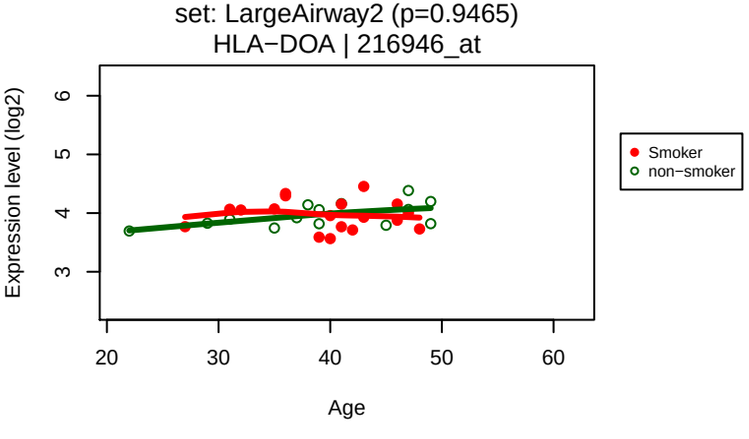

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DOA | 216946_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0946
p=0.026
p=0.0067
p=0.8292
p=0.9465
p=0.9628
p=0.5679
p=0.3041
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

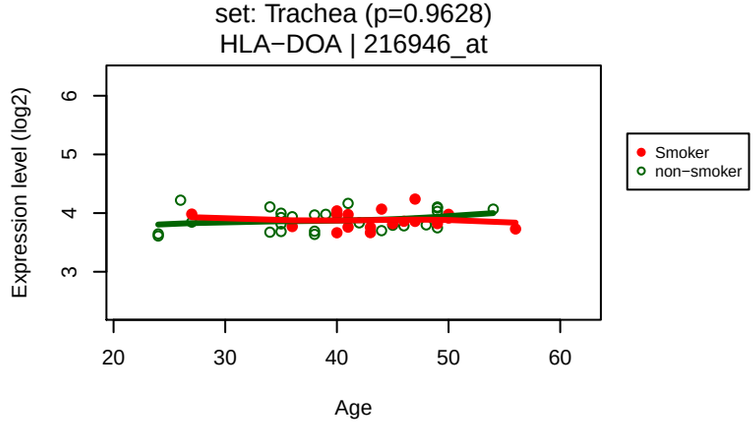

Gene expression ratio in Males
gene: HLA−DOA | 216946_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1431
p=0.8602
p=0.3844
p=0.2457
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DOA | 216946_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2137
p=0.3986
p=0.2698
p=0.944
SmallAirway2
LargeAirway2
Trachea
Macrophage2

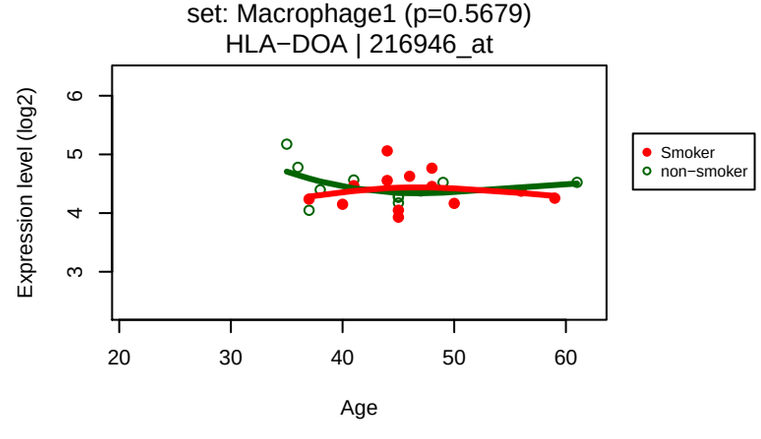

set: SmallAirway1 (p=0.0946)
HLA−DOA | 216946_at
Expression level (log2)
Age
Smoker
non−smoker

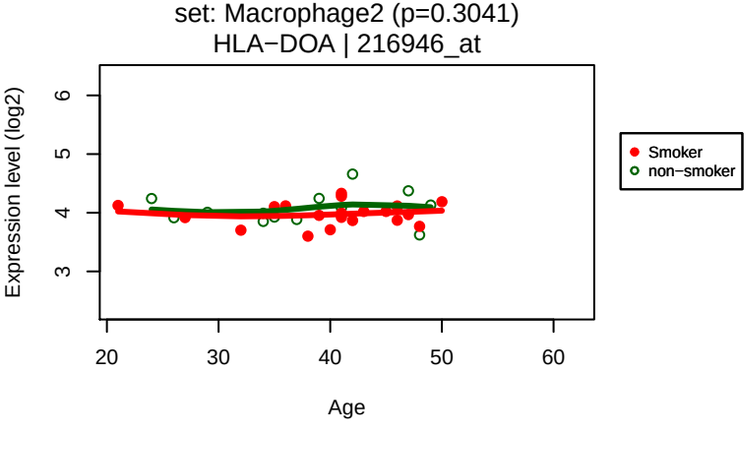

set: SmallAirway2 (p=0.026)
HLA−DOA | 216946_at
Expression level (log2)
Age
Smoker
non−smoker

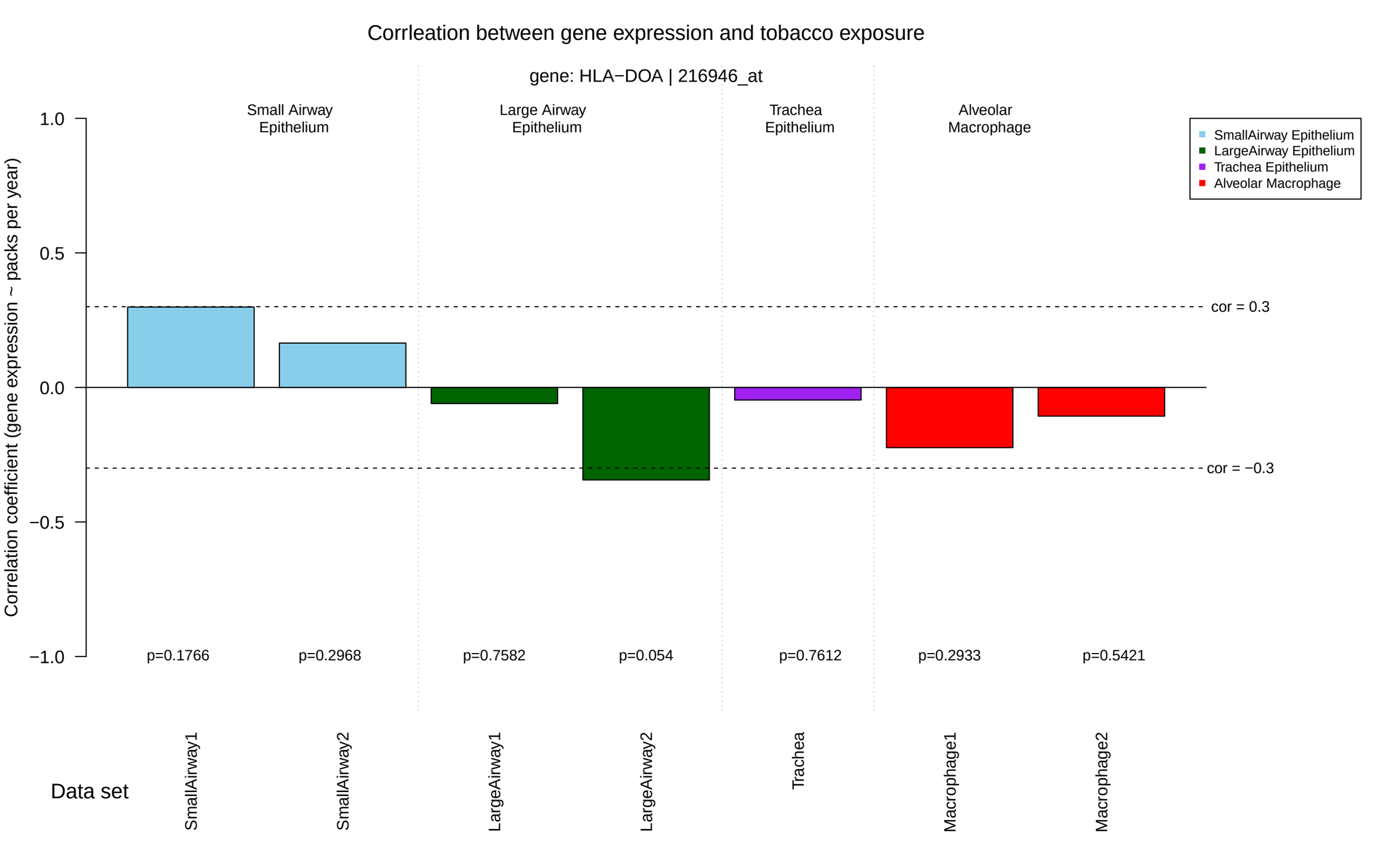

set: LargeAirway1 (p=0.8292)
HLA−DOA | 216946_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9465)
HLA−DOA | 216946_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9628)
HLA−DOA | 216946_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5679)
HLA−DOA | 216946_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3041)
HLA−DOA | 216946_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DOA | 216946_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1766
p=0.2968
p=0.7582
p=0.054
p=0.7612
p=0.2933
p=0.5421
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2